

Labwork 4: Threads

Nguyen Nhu Duy

1 Implementation

The grayscale CUDA implementation from Labwork 3 was improved by using 2D blocks and a 2D grid. Each CUDA thread directly processes one image pixel using its (x, y) coordinates, instead of using a 1D index.

2 2D Block Size Experiment

Different 2D block sizes were tested, including $(8, 8)$, $(16, 16)$, $(32, 16)$, $(16, 32)$, and $(32, 32)$. The execution time was measured using `time.time()`. A warm-up run was performed before timing, and the reported execution time was calculated as the average of five runs.

The 2D implementation achieved average execution times from approximately 0.000156 s to 0.000256 s. Compared with the previous 1D implementation, which required approximately 0.027–0.030 s, the 2D implementation was about 105–192 times faster. The best result was obtained with a $(16, 16)$ block size at approximately 0.000156 s.

